VISUALIZACION Y ***EXPORTACIONES*** de reconstrucciones COLMAP

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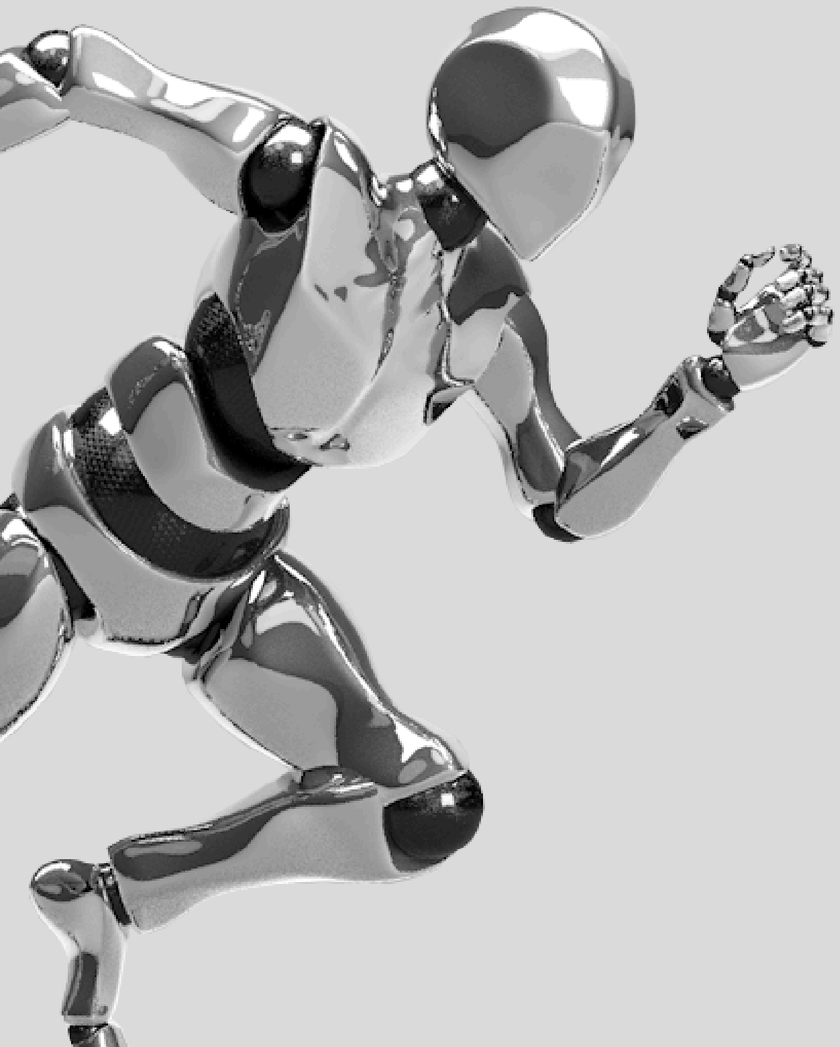
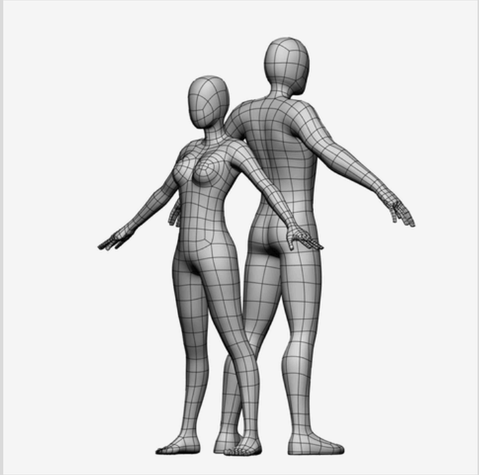


ÍNDICE

ARCHIVOS EXPORTABLES DESDE COLMAP

- COLMAP genera reconstrucciones 3D a partir de imágenes.
- Exporta nubes de puntos y mallas en varios formatos:

.PLY: NUBE DE PUNTOS O MALLA DENSA.
.OBJ: MALLA CON TEXTURA.
.FBX: USADO EN ENTORNOS COMO UNITY.



QUÉ RESULTADOS GENERA COLMAP?

PARÁMETROS DE RECONSTRUCCIÓN EN COLMAP

Match features:

- Extracción: SIFT (por defecto).
- Emparejamiento: exhaustive, sequential, spatial, etc.

Structure-from-Motion:

- Estimación de cámaras: Pinhole, Simple Radial, etc.
- Bundle Adjustment para optimizar cámaras y puntos 3D.
- Multi

Multi-view Stereo (MVS):

- Genera mapas de profundidad.
- Fusión produce fused.ply (nube densa).

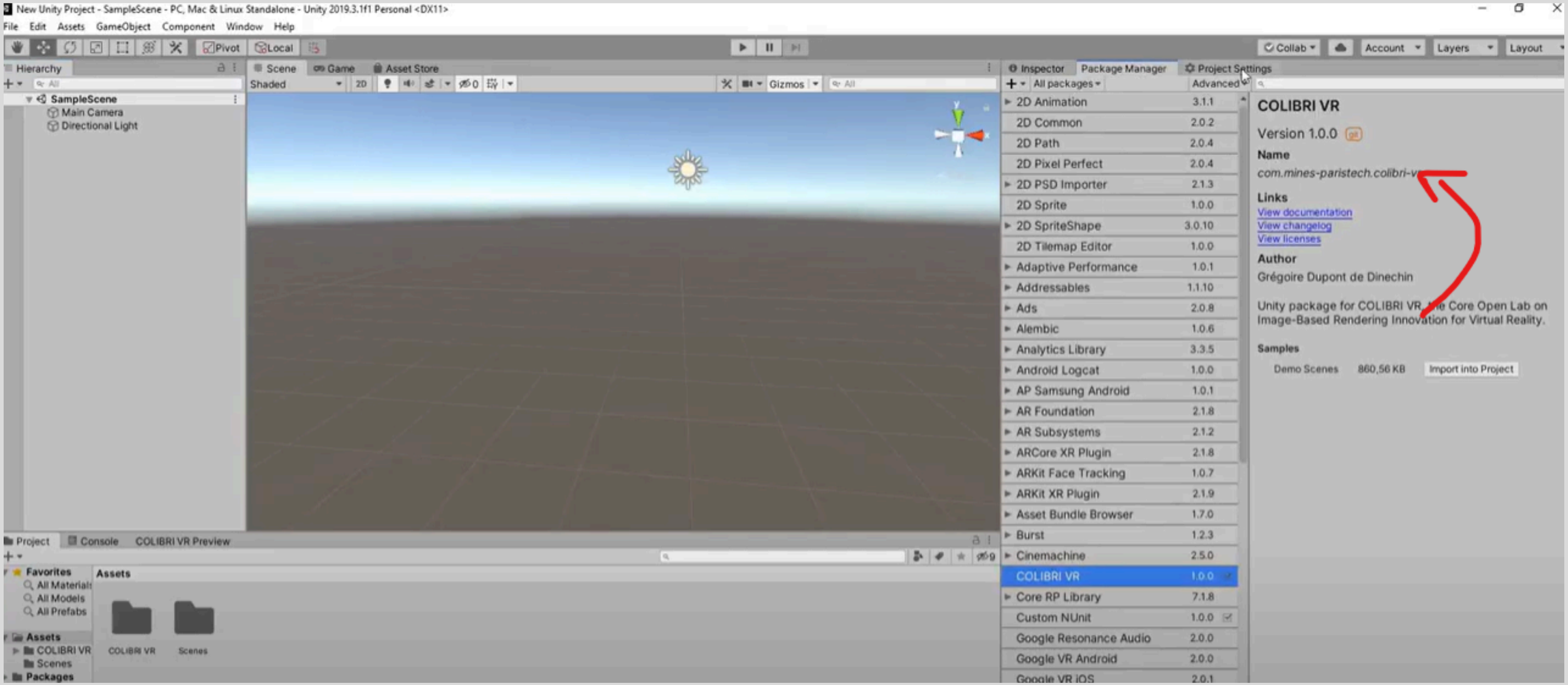
¿CÓMO EXPORTAR?

1. Completa la reconstrucción.
2. En el GUI: File > Export model
    - Elegir: Formato: .ply, .obj, .fbx
    - Tipo: sparse, dense, mesh

```
colmap model_converter \
    --input_path ./sparse/0 \
    --output_path ./output \
    --output_type PLY
```

1. A veces se requiere convertir el archivo .ply a .obj o .fbx:
    -
```
import open3d as o3d
pcd = o3d.io.read_point_cloud("modelo.ply")
o3d.io.write_point_cloud("modelo.obj", pcd)
```

EJEMPLOS ***PRÁCTICOS CON UNITY***


New Unity Project - SampleScene - PC, Mac & Linux Standalone - Unity 2019.3.1f1 Personal <DX11>
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2D Path 2.0.4
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AP Samsung Android 1.0.1
AR Foundation 2.1.8
AR Subsystems 2.1.2
ARCore XR Plugin 2.1.8
ARKit Face Tracking 1.0.7
ARKit XR Plugin 2.1.9
Asset Bundle Browser 1.7.0
Burst 1.2.3
Cinemachine 2.5.0
COLIBRI VR 1.0.0
Core RP Library 7.1.8
Custom NUnit 1.0.0
Google Resonance Audio 2.0.0
Google VR Android 2.0.0
Google VR iOS 2.0.1
COLIBRI VR
Version 1.0.0
Name
com.mines-paristech.colibri-vr
Links
View documentation
View changelog
View licenses
Author
Grégoire Dupont de Dinechin
Unity package for COLIBRI VR. The Core Open Lab on Image-Based Rendering Innovation for Virtual Reality.
Samples
Demo Scenes 860,56 KB Import into Project
Project Console COLIBRI VR Preview
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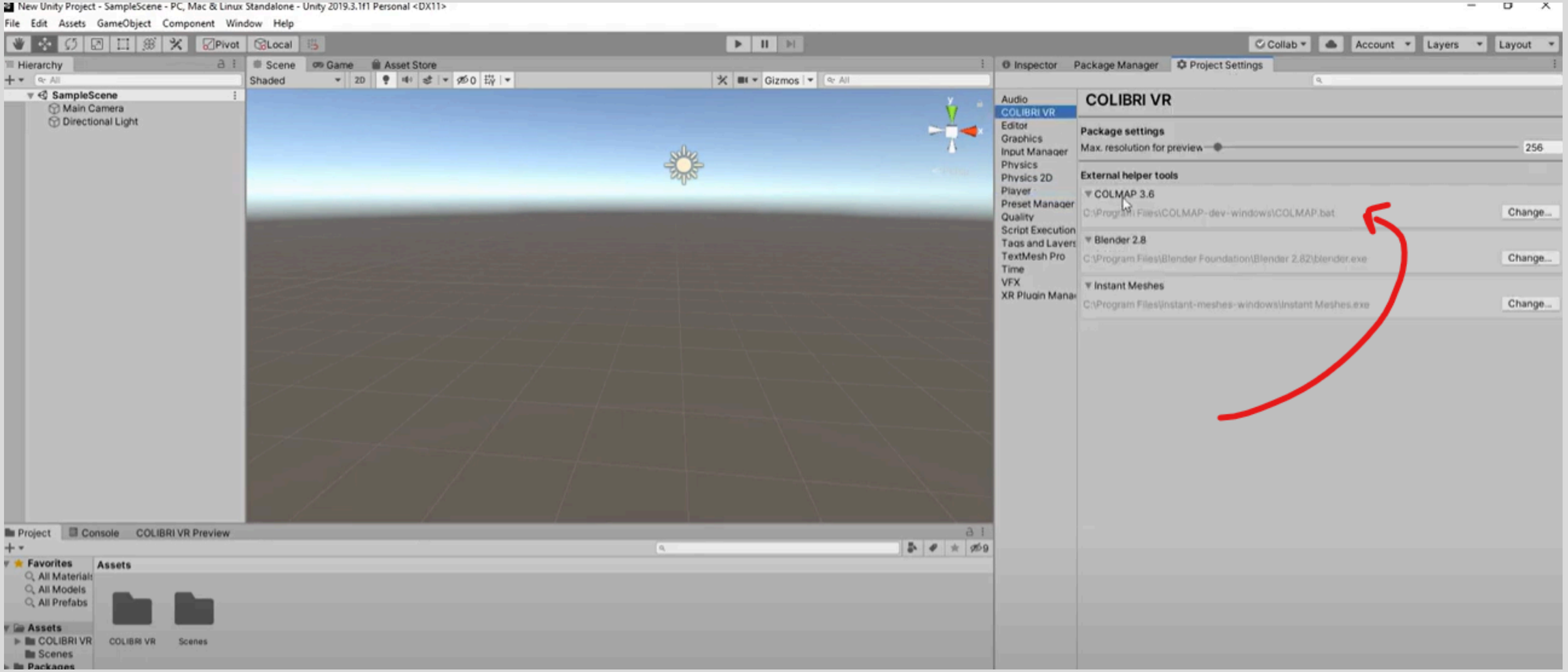
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C:\Program Files\Blender Foundation\Blender 2.82\blender.exe
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Instant Meshes
C:\Program Files\Instant-meshes-windows\Instant Meshes.exe
Change...
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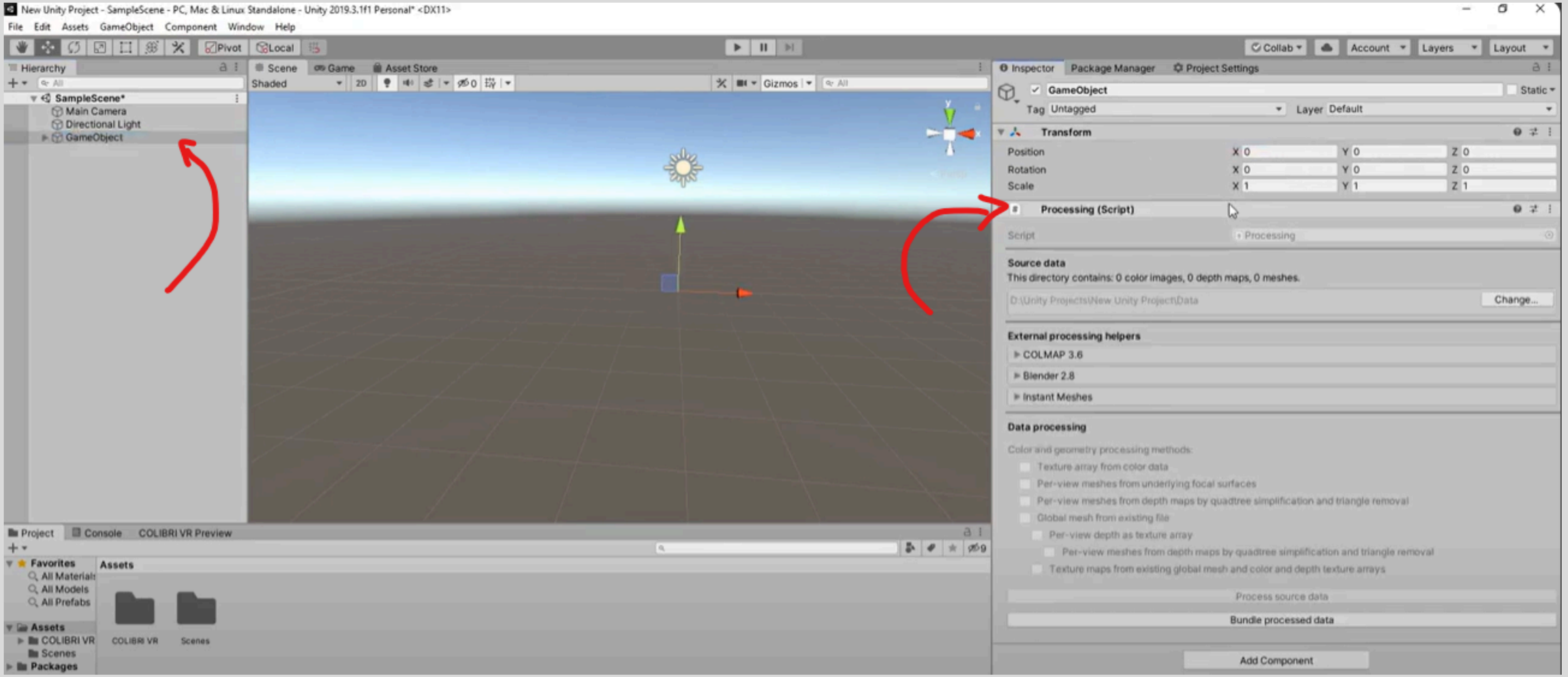
New Unity Project - SampleScene - PC, Mac & Linux Standalone - Unity 2019.3.1f1 Personal* <DX11>
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Source data
This directory contains: 0 color images, 0 depth maps, 0 meshes.
D:\Unity Projects\New Unity Project\Data
Change...
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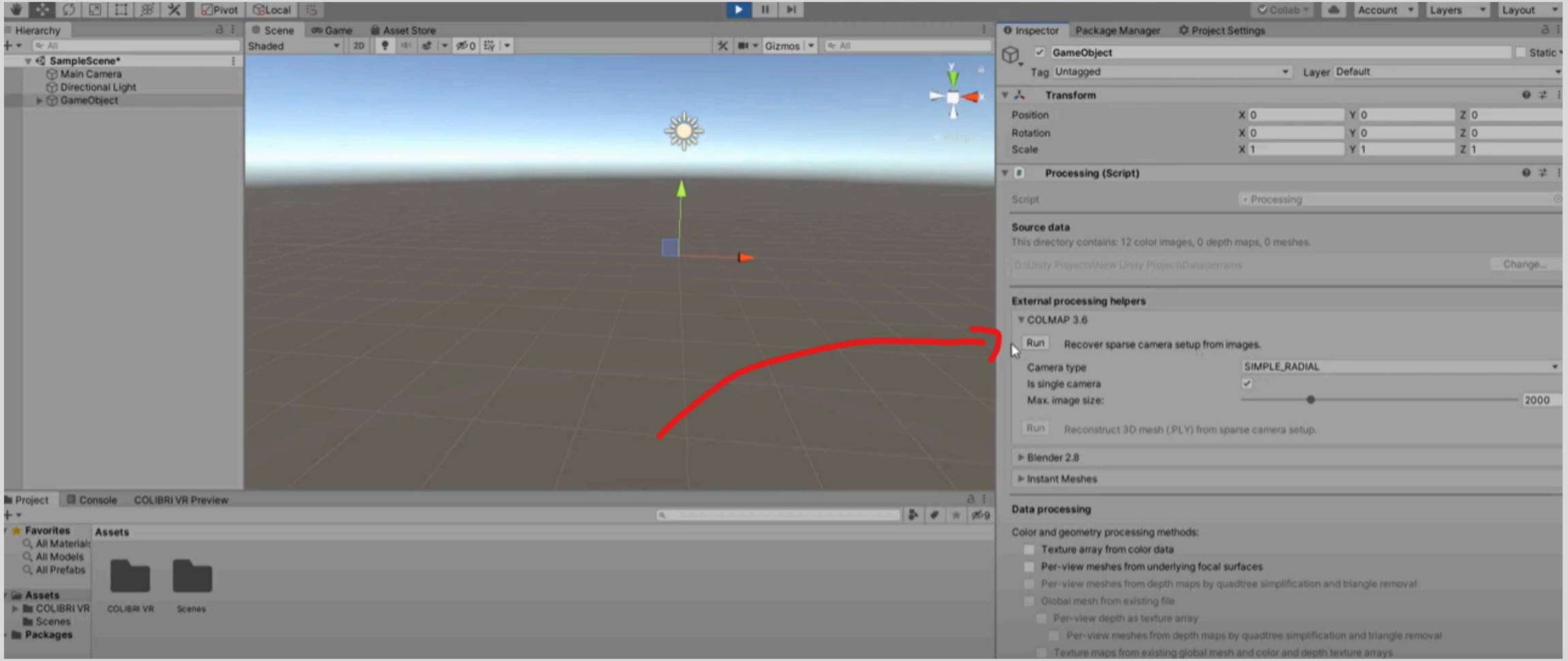
las posiciones relativas, las orientaciones y los parámetros de la cámara de la configuración de captura en función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Processing (Script)
Script
Processing
Source data
This directory contains: 12 color images, 0 depth maps, 0 meshes.
Change...
External processing helpers
COLMAP 3.6
Run
Recover sparse camera setup from images.
Camera type
SIMPLE_RADIAL
Is single camera
Max. image size:
2000
Reconstruct 3D mesh (.PLY) from sparse camera setup.
Blender 2.8
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Per-view meshes from depth maps by quadtree simplification and triangle removal
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Texture maps from existing global mesh and color and depth texture arrays
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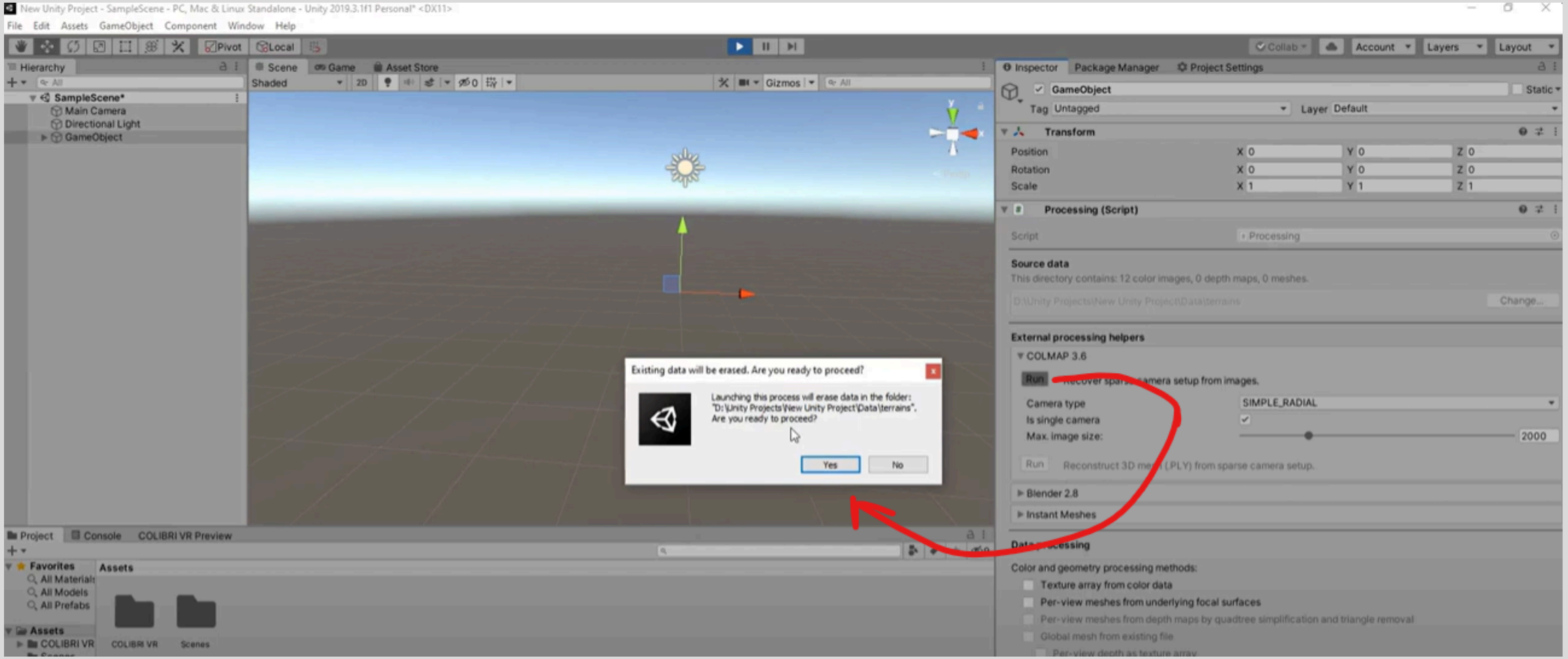
New Unity Project - SampleScene - PC, Mac & Linux Standalone - Unity 2019.3.1f1 Personal* <DX11>
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Tag Untagged Layer Default
Transform
Position X 0 Y 0 Z 0
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Processing (Script)
Script Processing
Source data
This directory contains: 12 color images, 0 depth maps, 0 meshes.
Change...
External processing helpers
COLMAP 3.6
Run
Camera type SIMPLE_RADIAL
Is single camera
Max. image size: 2000
Run Reconstruct 3D mesh (PLY) from sparse camera setup.
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Color and geometry processing methods:
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Global mesh from existing file
Existing data will be erased. Are you ready to proceed?
Launching this process will erase data in the folder: "D:\Unity Projects\New Unity Project\Data\terrains". Are you ready to proceed?
Yes
No
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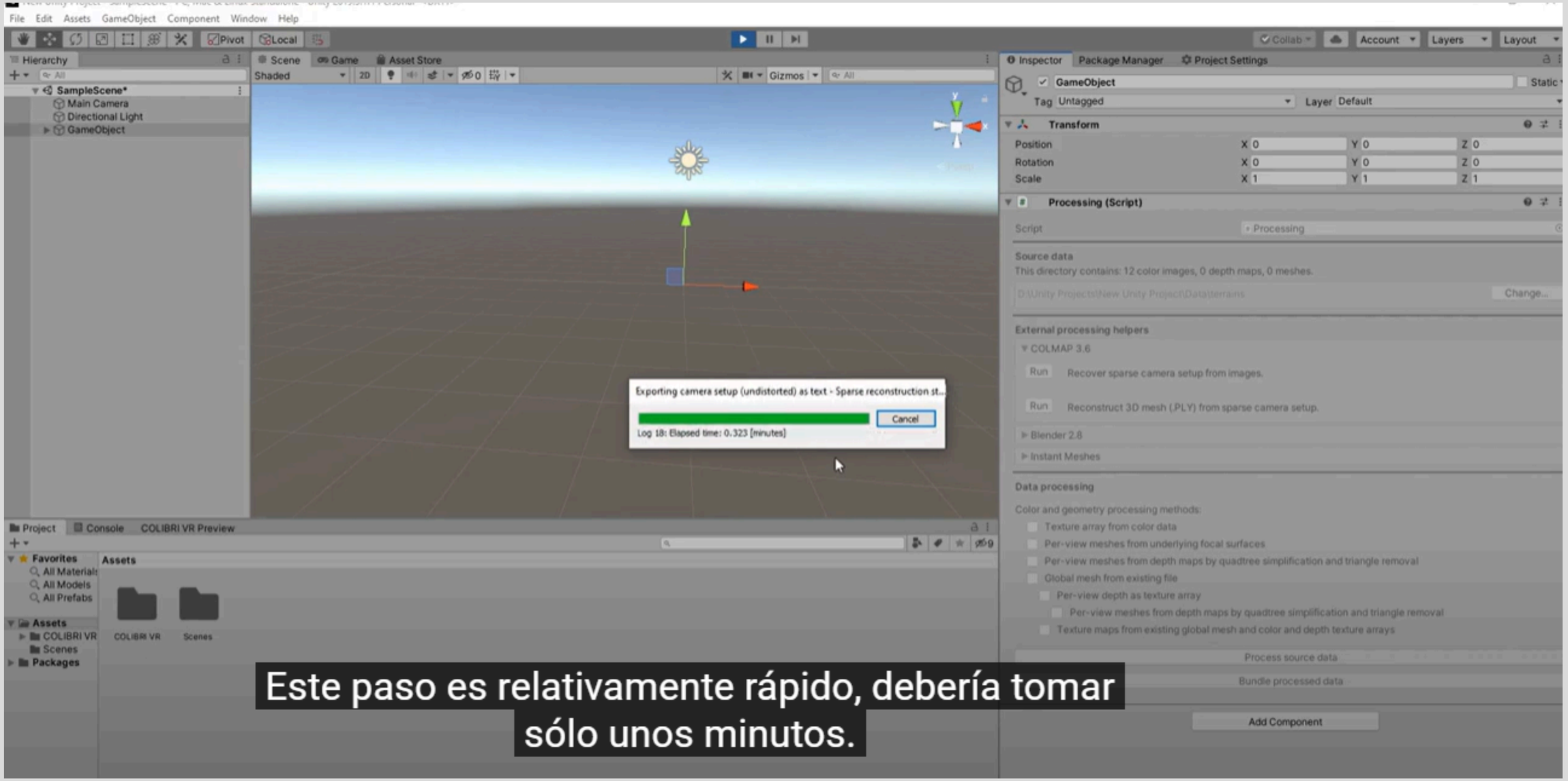
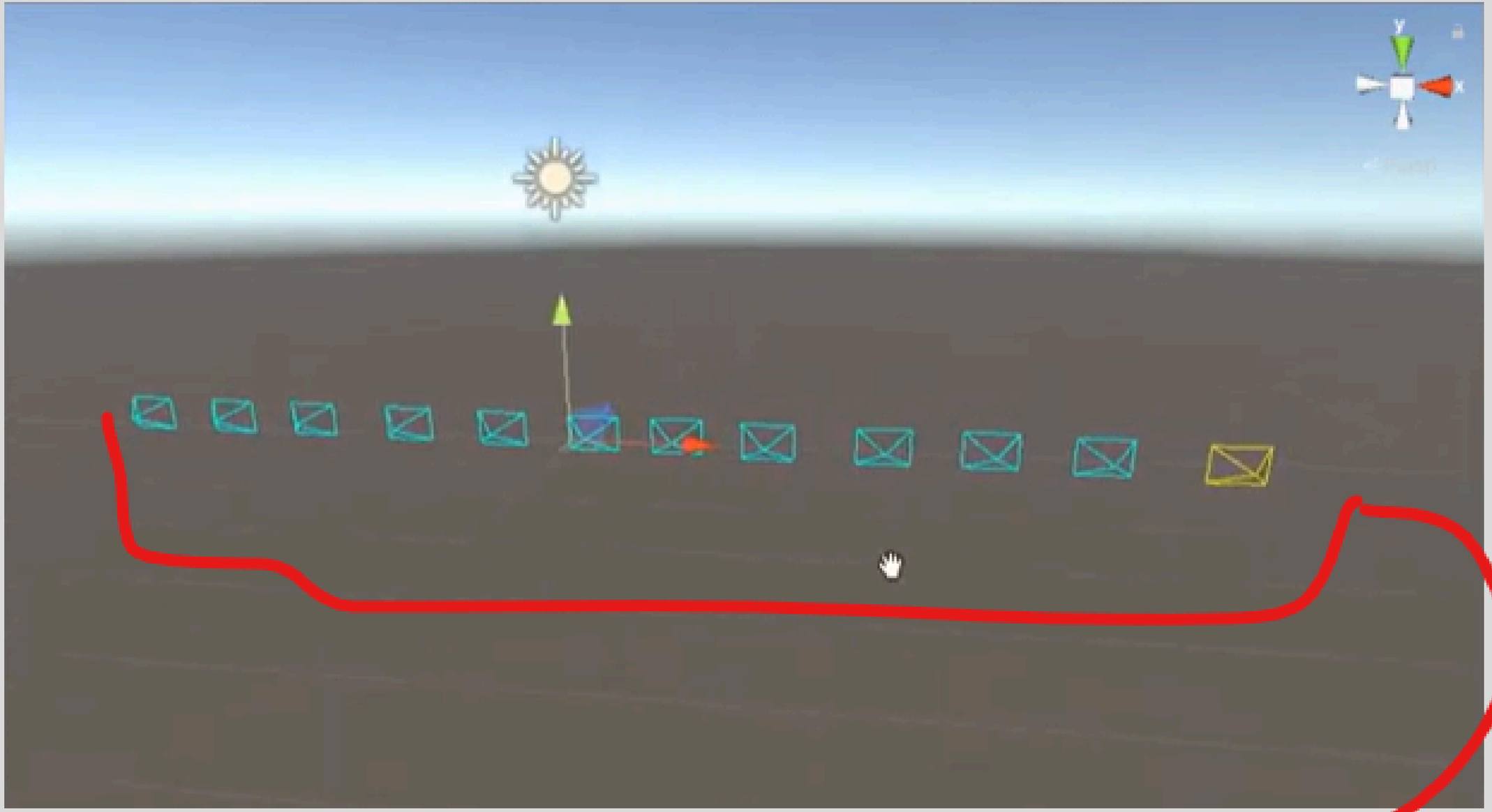
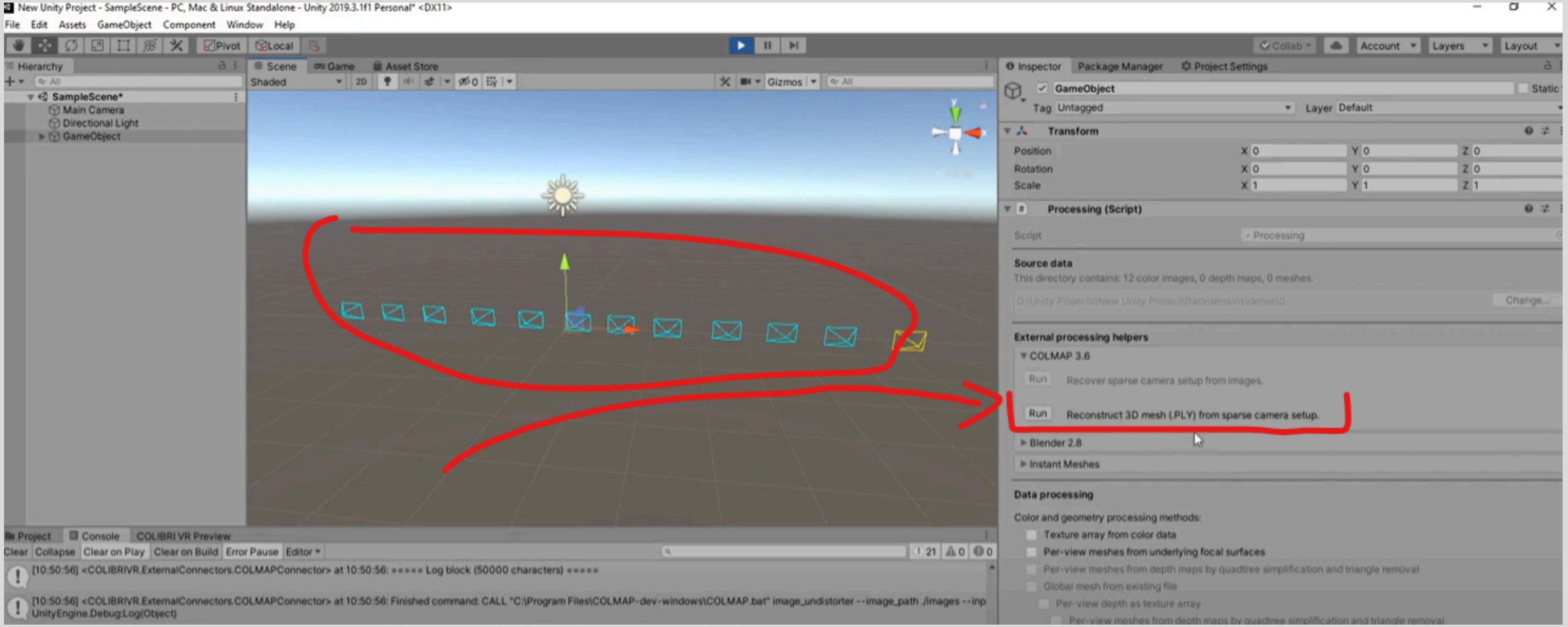


Este paso es relativamente rápido, debería tomar sólo unos minutos.


y
x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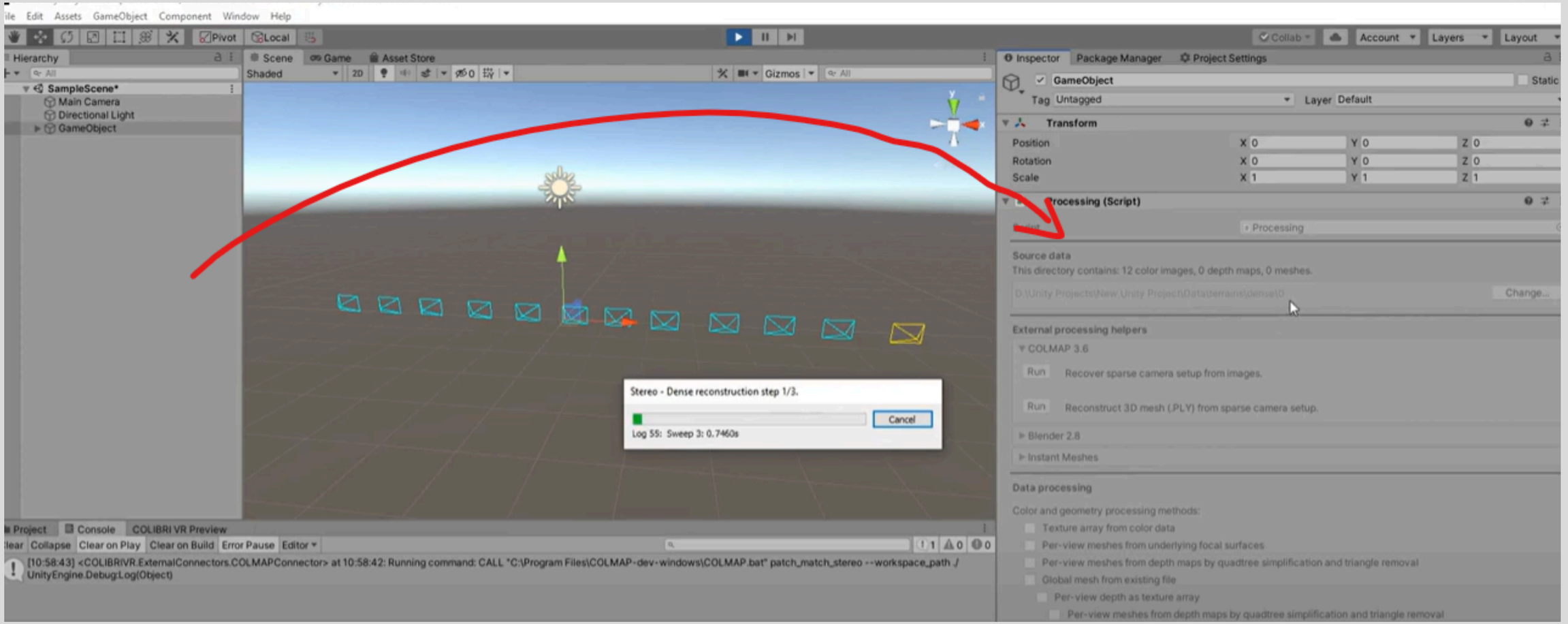
[10:50:56] <COLIBRIVR.ExternalConnectors.COLMAPConnector> at 10:50:56: ===== Log block (50000 characters) =====
[10:50:56] <COLIBRIVR.ExternalConnectors.COLMAPConnector> at 10:50:56: Finished command: CALL "C:\Program Files\COLMAP-dev-windows\COLMAP.bat" image_undistorter --image_path ./images --inp
UnityEngine.Debug:Log(Object)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Stereo - Dense reconstruction step 1/3.
Cancel
Log 55: Sweep 3: 0.7460s
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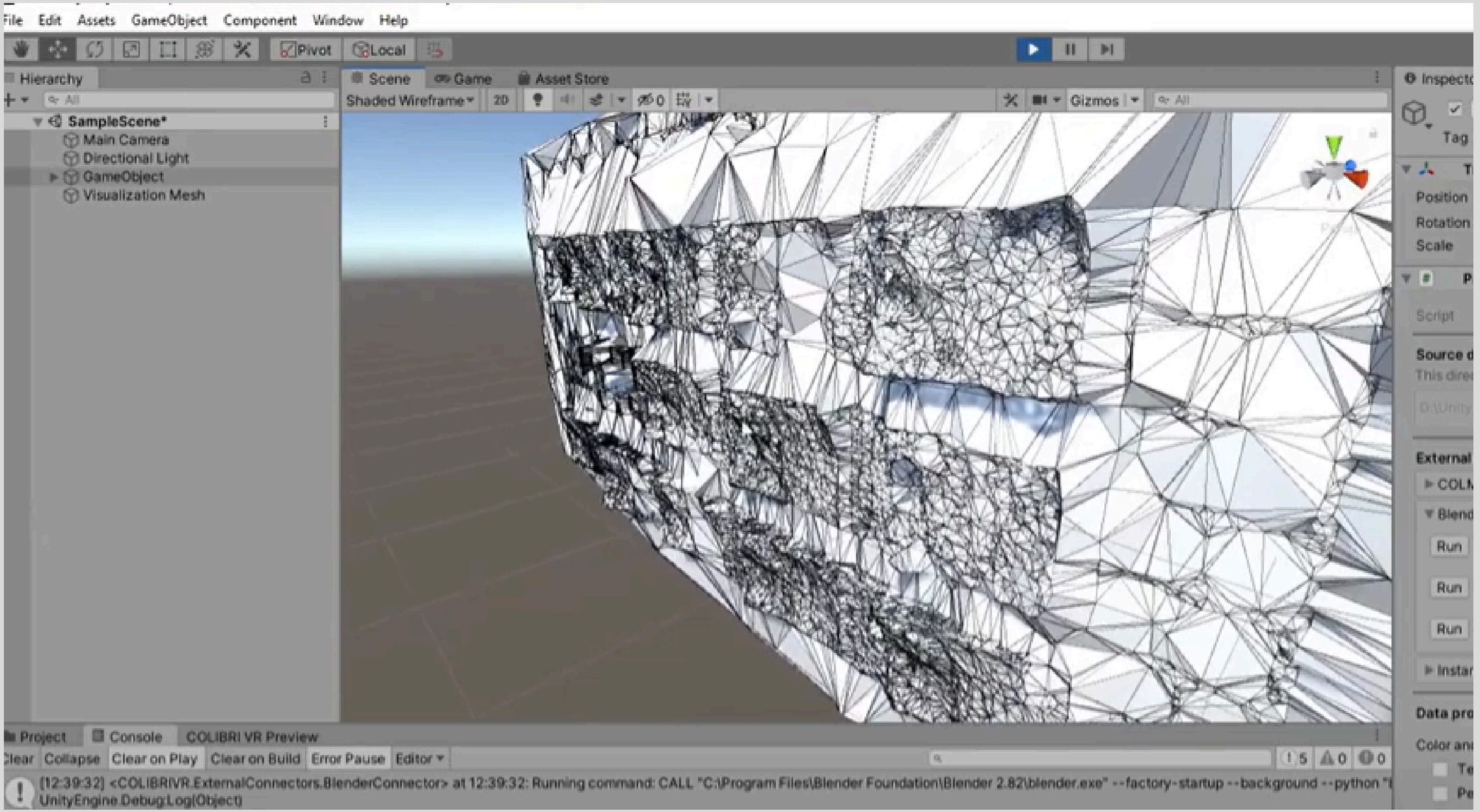
[10:58:43] <COLIBRIVR.ExternalConnectors.COLMAPConnector> at 10:58:42: Running command: CALL "C:\Program Files\COLMAP-dev-windows\COLMAP.bat" patch_match_stereo --workspace_path ./
UnityEngine.Debug:Log(Object)

Bien, ahora tenemos un archivo "meshed-delaunay.obj"
que podemos usar en Unity.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Run
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Clear on Play
Clear on Build
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[12:39:32] <COLIBRIVR.ExternalConnectors.BlenderConnector> at 12:39:32: Running command: CALL "C:\Program Files\Blender Foundation\Blender 2.82\blender.exe" --factory-startup --background --python
UnityEngine.Debug:Log(Object)

UNA VEZ QUE COLMAP HA GENERADO UNA RECONSTRUCCIÓN 3D, YA SEA UNA NUBE DE PUNTOS O UNA MALLA DENSA, EL SIGUIENTE PASO ES PREPARAR ESOS DATOS PARA SU USO EN ENTORNOS DE VISUALIZACIÓN COMO UNITY, THREE.JS O BLENDER. PARA ELLO, ES NECESARIO EXPORTAR, AJUSTAR Y LIMPIAR ADECUADAMENTE LOS MODELOS.

Exportación desde COLMAP

COLMAP permite exportar los resultados en varios formatos estándar o mediante herramientas externas en .fbx. Estos formatos son ampliamente compatibles con aplicaciones 3D modernas. Al exportar, es común utilizar:

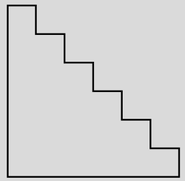
.ply para nubes de puntos crudas o filtradas.
.obj para mallas texturizadas generadas con Delaunay o Poisson.
.fbx si se requiere compatibilidad directa con motores como Unity.

AJUSTE DE ESCALA, ORIENTACIÓN Y LIMPIEZA (THREE.JS)

1. Cargar la nube de puntos o malla

```
import { GLTFLoader } from 'three/examples/jsm/loaders/GLTFLoader';
import { PointLight, Scene, PerspectiveCamera, WebGLRenderer } from 'three';

const loader = new GLTFLoader();
const scene = new Scene();

loader.load('/models/colmap_output.glb', (gltf) => {
  const model = gltf.scene;

  // Ajustes clave:
  model.scale.set(0.01, 0.01, 0.01); // ⬅️ Ajusta la escala
  model.rotation.set(-Math.PI / 2, 0, 0); // ⬅️ Orientación (si Z apunta arriba)
  model.position.set(0, 0, 0); // ⬅️ Posición inicial

  scene.add(model);
});
```

Cuando se exporta desde COLMAP (.ply, .obj), se puede obtener ruido, duplicados o geometría sobrante. Para integrar esto correctamente en una app 3D, es útil filtrar o limpiar antes o durante la carga.



¿Qué es "limpieza"?

1. Eliminar puntos o geometría fuera de un rango espacial (por ejemplo, demasiado alejados).
2. Filtrar por bounding box: sólo conservar una región 3D.
3. Ocultar mallas específicas si son subcomponentes innecesarios.

Opcion 1: Limpieza con bounding box en Three.js (si ya está cargado)

```
model.traverse((child) => {
  if (child.isMesh) {
    child.geometry.computeBoundingBox();
    const box = child.geometry.boundingBox;

    const min = box.min;
    const max = box.max;

    // Si el objeto está fuera de un rango aceptable, lo ocultamos
    const isOutside = max.z > 5 || min.z < -2 || max.x > 5 || min.x < -5;
    if (isOutside) {
      child.visible = false;
    }
  }
});
```

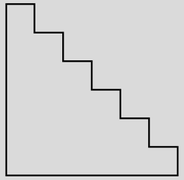
Esto permite en tiempo real desactivar parte de la geometría cargada

Opción 2: Limpieza previa con herramientas gráficas

En CloudCompare:
Cargar .ply → Usar "Crop Tool" para seleccionar una región 3D.
Aplicar filtro por densidad o distancia.
Exportar la versión reducida.

En Blender:
Importar .ply o .obj.
Usar modo Edit → Box select para borrar puntos aislados.
Exportar como .glb o .gltf limpio.

Opción 3: Limpieza automatizada por densidad (en Python)
Si se trabaja por fuera de Three.js, se puede hacer la limpieza con Open3D:



```python
import open3d as o3d

pcd = o3d.io.read_point_cloud("dense.ply")
cl, ind = pcd.remove_statistical_outlier(nb_neighbors=20, std_ratio=2.0)

pcd_clean = pcd.select_by_index(ind)
o3d.io.write_point_cloud("cleaned_dense.ply", pcd_clean)
```

Esto elimina puntos dispersos basados en la densidad local.

GRACIAS POR LA ATENCIÓN